Practice workbook · Turning sums of trig terms into products, and products into sums

Adding or subtracting two addition formulae to turn a product of trig functions into a sum, or a sum into a product.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-22

1. Question 10 · part b · 3 marks

2020 NOV · Paper 02

Worked-example retry · Attempt part b only.

10 (a) Show that

$$\cos(A + B) + \cos(A - B) = 2\cos A \cos B$$

(2)

(b) Hence show that

$$\cos P + \cos Q = 2\cos\frac{P + Q}{2}\cos\frac{P - Q}{2}$$

(3)

(c) Solve, for $0 \leqslant \theta \leqslant \frac{\pi}{2}$, the equation

$$\cos 5\theta + \cos 7\theta = 0$$

Give each solution in terms of π

(4)

(d) Show that

$$\cos 8x + 2\cos 6x + \cos 4x = 4\cos 6x \cos^2 x$$

(3)

(e) Use calculus to find the exact value of

$$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx$$

(4)

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1. Question 10 · part b · 3 marks · Working and answer

Working and answer

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2. Question 10 · part a · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

10 (a) Show that

$$\cos(A + B) + \cos(A - B) = 2\cos A \cos B$$

(2)

(b) Hence show that

$$\cos P + \cos Q = 2\cos\frac{P + Q}{2}\cos\frac{P - Q}{2}$$

(3)

(c) Solve, for $0 \leqslant \theta \leqslant \frac{\pi}{2}$, the equation

$$\cos 5\theta + \cos 7\theta = 0$$

Give each solution in terms of π

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(4)

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2. Question 10 · part a · 2 marks · Working and answer

Working and answer

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3. Question 10 · part d · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part d only.

10 (a) Show that

$$\cos(A + B) + \cos(A - B) = 2\cos A \cos B$$

(2)

(b) Hence show that

$$\cos P + \cos Q = 2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2}$$

(3)

(c) Solve, for $0 \leqslant \theta \leqslant \frac{\pi}{2}$, the equation

$$\cos 5\theta + \cos 7\theta = 0$$

Give each solution in terms of π

(4)

(d) Show that

$$\cos 8x + 2\cos 6x + \cos 4x = 4\cos 6x \cos^2 x$$

(3)

(e) Use calculus to find the exact value of

$$\int_0^{\frac{\pi}{3}} \cos 6x \cos^2 x \, dx$$

(4)

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3. Question 10 · part d · 3 marks · Working and answer

Working and answer

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4. Question 10 · part a · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

10 Using suitable results for $\sin(A + B)$ and $\sin(A - B)$ from the formulae page,

(a) show that $2\sin 4x \cos x = \sin 5x + \sin 3x$

(3)

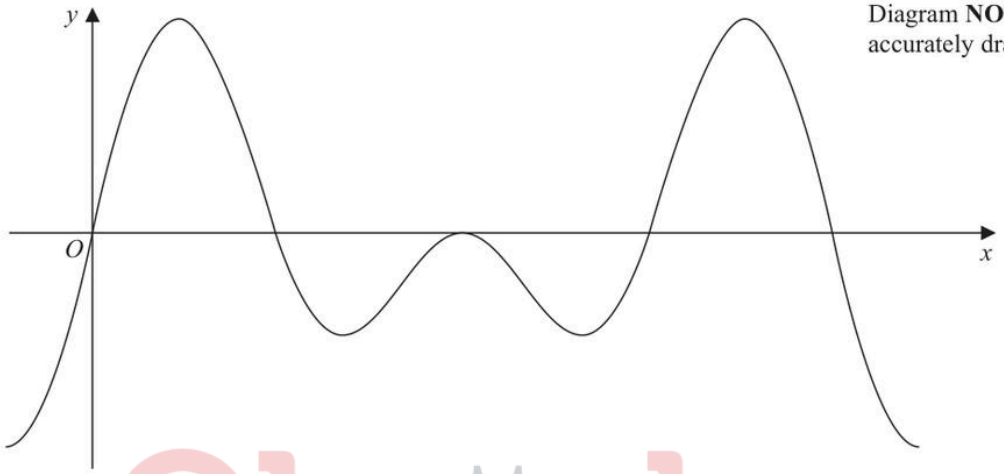


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows part of a sketch of the curve $y = 6\sin 4x \cos x$

(b) Using calculus, find the total area bounded by the curve and the x-axis between $x = 0$ and $x = \frac{\pi}{2}$

Give your answer to 3 significant figures.

(8)

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P 7 1 6 4 1 A 0 2 8 3 6

4. Question 10 · part a · 3 marks · Working and answer

Working and answer

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5. Question 6 · part a · 2 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

6 (a) Show that

$$\sin(A + B) + \sin(A - B) = 2\sin A \cos B$$

(2)

(b) Hence express $2\sin 7x \cos x$ in the form $\sin mx + \sin nx$ where m and n are integers, giving the value of m and the value of n.

(1)

(c) Use calculus to evaluate

$$\int_0^{\frac{\pi}{4}} (6\sin 7x \cos x)\,dx$$

(4)

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5. Question 6 · part a · 2 marks · Working and answer

Working and answer

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6. Question 6 · part b · 1 mark

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part b only.

6 (a) Show that

$$\sin(A + B) + \sin(A - B) = 2\sin A \cos B$$

(2)

(b) Hence express $2\sin 7x \cos x$ in the form $\sin mx + \sin nx$ where m and n are integers, giving the value of m and the value of n.

(1)

(c) Use calculus to evaluate

$$\int_0^{\frac{\pi}{4}} (6\sin 7x \cos x)\,dx$$

(4)

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6. Question 6 · part b · 1 mark · Working and answer

Working and answer

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7. Question 10 · part a · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

10 Using formulae from page 2

(a) show that $\cos(A - B) - \cos(A + B) = 2\sin A \sin B$

(2)

(b) Hence show that $\cos 5\theta - \cos 9\theta = 2\sin 7\theta \sin 2\theta$

(1)

(c) Solve the equation

$$\cos 5\theta - \cos 9\theta = \sqrt{3}\sin 7\theta \quad \text{for} \quad 0 < \theta \leqslant \frac{1}{3}\pi$$

Give your solutions in terms of π

(7)

(d) Using calculus and showing your working, evaluate

$$\int_0^{\frac{\pi}{7}} 8\sin 7x \cos 2x \tan 2x \, dx$$

Give your answer to 3 decimal places.

(6)

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7. Question 10 · part a · 2 marks · Working and answer

Working and answer

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8. Question 10 · part b · 1 mark

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

10 Using formulae from page 2

(a) show that $\cos(A - B) - \cos(A + B) = 2\sin A \sin B$

(2)

(b) Hence show that $\cos 5\theta - \cos 9\theta = 2\sin 7\theta \sin 2\theta$

(1)

(c) Solve the equation

$$\cos 5\theta - \cos 9\theta = \sqrt{3}\sin 7\theta \quad \text{for} \quad 0 < \theta \leqslant \frac{1}{3}\pi$$

Give your solutions in terms of π

(7)

(d) Using calculus and showing your working, evaluate

$$\int_0^{\frac{\pi}{7}} 8\sin 7x \cos 2x \tan 2x \, dx$$

Give your answer to 3 decimal places.

(6)

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8. Question 10 · part b · 1 mark · Working and answer

Working and answer

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